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Supplementary Materials for

Selective conversion of syngas to light olefins

Feng Jiao, Jinjing Li, Xiulian Pan,* Jianping Xiao, Haobo Li, Hao Ma, Mingming Wei, Yang Pan, Zhongyue Zhou, Mingrun Li, Shu Miao, Jian Li, Yifeng Zhu, Dong Xiao, Ting He, Junhao Yang, Fei Qi, Qiang Fu, Xinhe Bao*

*Corresponding author. E-mail: panxl@dicp.ac.cn (X.P.); xhbao@dicp.ac.cn (X.B.)

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Materials and Methods

Catalyst preparation

$ZnCrO_x$ oxide was prepared by co-precipitation method. Briefly, 14.56 g $Zn(NO_3)_2 \cdot 6H_2O$, 5.6 g $Cr(NO_3)_3 \cdot 9H_2O$, 5.25 g $Al(NO_3)_3 \cdot 9H_2O$ were dissolved in 100 ml distilled water and $(NH_4)_2CO_3$ aqueous solution was used as the precipitant. Precipitation was allowed at 70 ℃, followed by aging for 3 h at the same temperature. After filtering and washing by distilled water, the resulting product was dried overnight at 110 ℃ and then calcined at 500 ℃ for 1 h.

Zeolite MSAPO was synthesized hydrothermally. Typically, 30% silica sol, 82% AlOOH, 85% phosphoric acid and triethylamine (TEA) were well dispersed in distilled water with a ratio of $SiO_2:Al_2O_3:H_3PO_4:TEA:H_2O$ = 0.48:1:1.6:3:50. Then the mixture was placed in a Teflon-lined autoclave, and kept for 12–28 h in a temperature range of 180−200 ℃. The resulting solid product was washed by centrifuging till pH of the supernate was 7.0−7.5. After drying for over 12 h at 110 ℃, it was calcined for 3 h at 550 ℃.

Commercial SAPO-34, ZSM-5 (Si/Al ratio = 80) and Beta zeolite (Si/Al ratio = 40) were purchased from Nankai University Catalyst Company, China.

Catalytic reaction tests

Catalytic reactions were performed in a continuous flow, fixed-bed stainless steel reactor furnished with a quartz lining. Typically 280–300 mg composite catalyst (20–40 mesh) with $ZnCrO_x$/MSAPO = 1/1 (mass ratio) was used unless otherwise stated. For a test with the oxide $ZnCrO_x$ without MSAPO as Mode 1 of Figure 2A, 140 mg catalyst was used. The catalyst was subjected to in situ reduction in H_2 for 2 h at 310 ℃ prior to reaction. Syngas with CO and H_2 contains 5% Ar as the internal standard for online gas chromatography (GC) analysis. Reaction was carried out under conditions of H_2/CO = 2.5, 2.5 MPa, 400 ℃ and 5143 ml/h $\cdot g_{cat}$ unless otherwise stated. Data were collected after at least 4 hours on stream.

A test with methanol as the feeding was also performed in a continuous flow, fixed-bed reactor at 400 ℃ and atmospheric pressure, over the composite catalyst and also just MSAPO. After reduction in H_2 at 310 ℃ for 2 h, methanol was introduced with He as the carrier gas. The weight hourly space velocity of methanol was set at 0.5 h^{-1}, equivalent to the yield of olefins of the OX-ZEO process assuming 100% methanol conversion.

Products were analyzed by an online GC (Agilent 7890B), which was equipped with a thermal conductivity detector (TCD) and a flame ionization detector (FID). Hayesep Q and 5Å molecular sieves packed columns were connected to TCD while HP-FFAP and HP-AL/S capillary columns were connected to FID. Oxygen-containing compounds and hydrocarbons up to C_{17} were analyzed by FID, while CO, CO_2, CH_4, C_2H_4, and C_2H_6 were analyzed by TCD. CH_4, C_2H_4 and C_2H_6 were taken as a reference bridge between FID and TCD. CO conversion was calculated on a carbon atom basis, i.e.

$$Con_{CO} = \frac{CO_{inlet} - CO_{outlet}}{CO_{inlet}} \times 100\%$$

Where CO_{inlet} and CO_{outlet} represent moles of CO at the inlet and outlet, respectively. CO_2 selectivity (Sel_{CO_2}) was calculated according to:

$$Sel_{CO_2} = \frac{CO_{2\,outlet}}{CO_{inlet} - CO_{outlet}}$$

Where $CO_{2\,outlet}$ denotes moles of CO_2 at the outlet.

The selectivity of individual hydrocarbon C_nH_m ($Sel_{C_nH_m}$) was obtained according to:

$$Sel_{C_nH_m} = \frac{nC_nH_{m\,outlet}}{\sum_1^n nC_nH_{m\,outlet}} \times 100\%$$

The selectivity to oxygenates was below 0.5%C and therefore was not reported in the product selectivity. The carbon balance was over 95%.

In situ near ambient pressure X-ray photoelectron spectroscopy (NAP-XPS)

In situ near ambient pressure X-ray photoelectron spectroscopy (NAP-XPS) was carried out at the beamline 9.3.2 at the Advanced Light Source, Berkeley. The catalyst was pressed into pellets, which were first degassed under ultrahigh vacuum (UHV) around 200 ℃ and then oxidized in 1×10^{-6} Torr O_2 to remove any carbon contamination at the catalyst surface. High-purity CO and H_2 were allowed into the analysis chamber with the partial pressure of CO up to 0.5 Torr and H_2 up to 0.15 Torr. Zn 3d, Cr 2p, O 1s, C 1s, Al 2s, and Si 2p spectra were in-situ acquired using the photon energy of 735 eV. The samples show a strong charging effect below 250 ℃. However, above 300 ℃ and in the presence of high pressure gases there is not much influence. Al 2s and Si 2p were used as the references to calibrate all binding energy positions.

Synchrotron-based VUV photoionization mass spectrometry (SVUV-PIMS)

Synchrotron-based VUV photoionization mass spectrometry (SVUV-PIMS) study was carried out at the combustion beamline of the National Synchrotron Radiation Laboratory at Hefei, China. As shown in fig. S9, a quartz reactor with a nozzle size of ~0.1 mm was designed, which was connected to the online SVUV-PIMS spectrometer. The detection limit of SVUV-PIMS is 0.1 ppm for CO_2. Atmospheric pressure syngas (H_2/CO = 2.5) was introduced in and the reaction was allowed in the temperature range of 355 – 400 ºC.

Ketene synthesis and catalytic conversion of ketene to light olefins

Ketene was synthesized via pyrolysis of acetic anhydride according to the previously reported procedure (24, 28). Control experiments were carried out to check the catalysis of MSAPO in ketene conversion to light olefins by feeding ketene directly to the MSAPO catalyst at 400 ℃ and the effluents were monitored by SVUV-PIMS.

Catalyst characterization

X-ray diffraction (XRD) was measured on a PANalytical X'pert PPR diffractometer equipped with CuKα radiation source (λ = 1.5418 Å), operated at 40 mA and 40 kV. XRD patterns were recorded in the range of 2θ = 5–80 º.

(S)TEM experiments were carried out on the FEI F30 transmission electron microscopes with a resolution of 0.17 nm operated at an acceleration voltage of 300 kV, and JEM-ARM200F TEM with a probe corrector working at 200 kV.

Temperature-programmed-desorption of NH_3 (NH_3–TPD) was performed on an Micromeritics AutoChem 2910 equipped with a thermal conductivity detector (TCD). The catalyst was first pretreated in a flowing Ar at 450 ℃ for 1.5 h. After cooling down to 100 ℃ under flowing Ar, the sample was exposed to 15% NH_3/He at 100 ℃. Then the sample was swept by He at 100 ℃ till a stable baseline was obtained. Subsequently, signal was recorded while the temperature was increased from 100 to 800 ℃ at a heating rate of 10 ℃/min.

Temperature-programmed-reduction of CO and the following surface reaction with H_2 was also performed with the same setup of SVUV-PIMS. After in situ H_2 reduction for 2 h at 310 ℃ followed by degasing to vacuum down to around (4–6) $\times 10^{-5}$ Pa at 25 ℃. CO was allowed in at a flow rate of 12.7 ml/min. Subsequently, the catalyst was heated to 400 ℃ at a rate of 10 ℃/min and held for 15 min at this temperature. The effluent was monitored at a photon energy hυ = 14.50 eV, which allowed detection of both CO and CO_2. Subsequently, the catalyst was cooled down to 25 ℃ and evacuated again. Then H_2 was introduced and the temperature was increased to 400 ℃ at the same rate. The effluent was monitored at hυ = 13.00 eV, which allowed detection of hydrocarbons including methane (ionization energy = 12.61 eV).

Nitrogen adsorption-desorption was carried out on a Quantachrome NOVA 4200e. Before analysis, all samples were degassed under vacuum at 300 ℃ for 6 h. Isotherms were recorded at 77 K.

Thermogravimetric Analysis (TGA) was conducted on Pyris Diamond TG-DTG in a flowing air (100 ml/min) and the heating rate was 10 ℃/min from 30 to 900 ℃.

<u>Theoretical calculations</u>

Theoretical calculations were carried out based on Density Functional Theory (DFT) framework by using Vienna ab initio simulation packages (VASP) with the projector augmented wave (PAW) scheme (29). The Perdew-Burke-Ernzerhof (PBE) functional was adopted for the exchange-correlation effect (30). A model of spinel $ZnCr_2O_4$ (111) surface with two repeating unit layers (12 atomic layers) and the vacuum thickness of about 20 Å were used to study all adsorption and other kinetic processes. The Brillouin zone was sampled at gamma. All atoms except the atomic layers in the bottom were relaxed until the forces are smaller than 0.05 eV/Å.

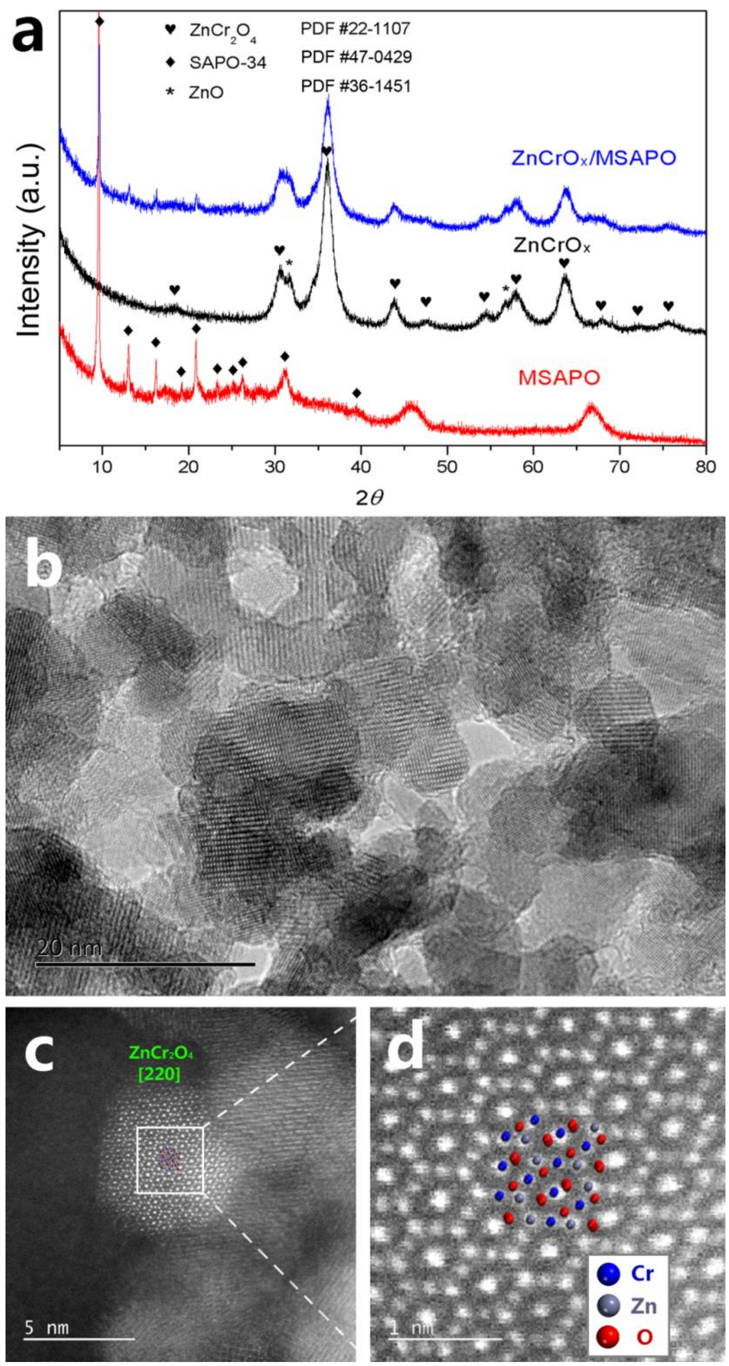

Fig. S1.(a) XRD patterns of $ZnCrO_x$, MSAPO and the resulting composite catalyst. $ZnCrO_x$ exhibits a typical spinel structure, corresponding to PDF #22-1107. The diffraction peaks of MSAPO matches with the characteristic diffraction of SAPO-34. (b)-(d) High resolution (S)TEM images of $ZnCrO_x$ at different regions of the specimen with (c) and (d) showing the atomic structure of spinel.

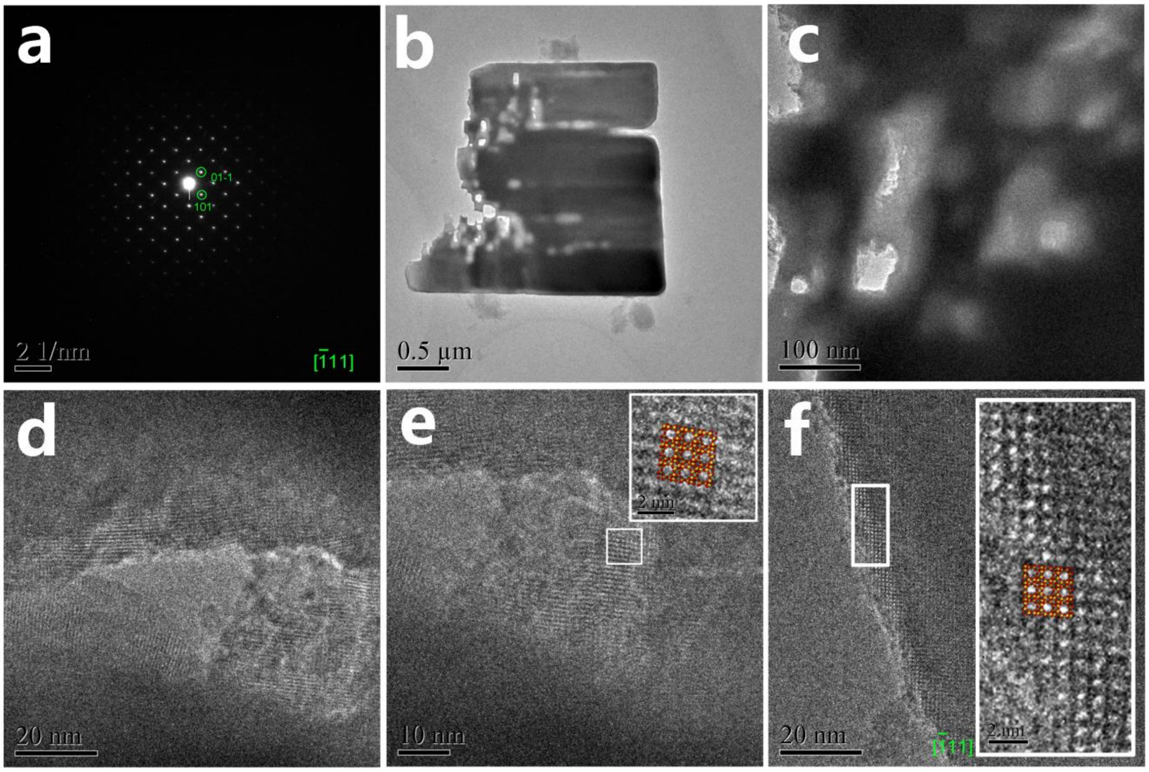

Fig. S2 TEM images of MSAPO. (a) Electron diffraction pattern showing typical SAPO-34 structure observed along the [-111] axis. (b)–(f) Typical morphology of MSAPO with (c)–(e) stepwise enlarged images, which exhibit clearly a hierarchical pore structure with macro-, meso- and micropores. The lattice structure is also clearly observed in (d)–(f) corresponding to SAPO-34. The insets in (e) and (f) show the enlarged image of the regions marked by the white rectangles and the pore structures fit very well with that of characteristic CHA crystal with micropores observed along the [-111] axis (denoted with yellow and red colors).

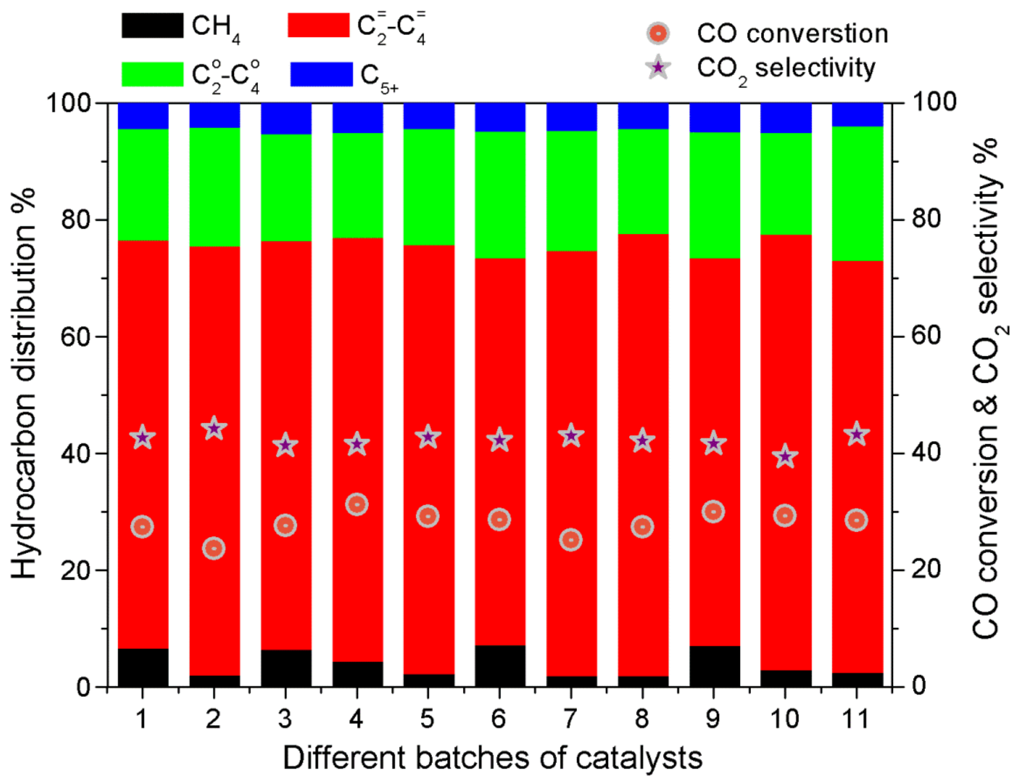

Fig. S3 Reproducibility test of catalysts, which have been prepared and evaluated in different batches under the same reaction conditions. The results demonstrate that the reactivity is well reproducible. Reaction conditions: 400 ℃, 2.5 MPa and GHSV 5143 ml/h g_{cat}.

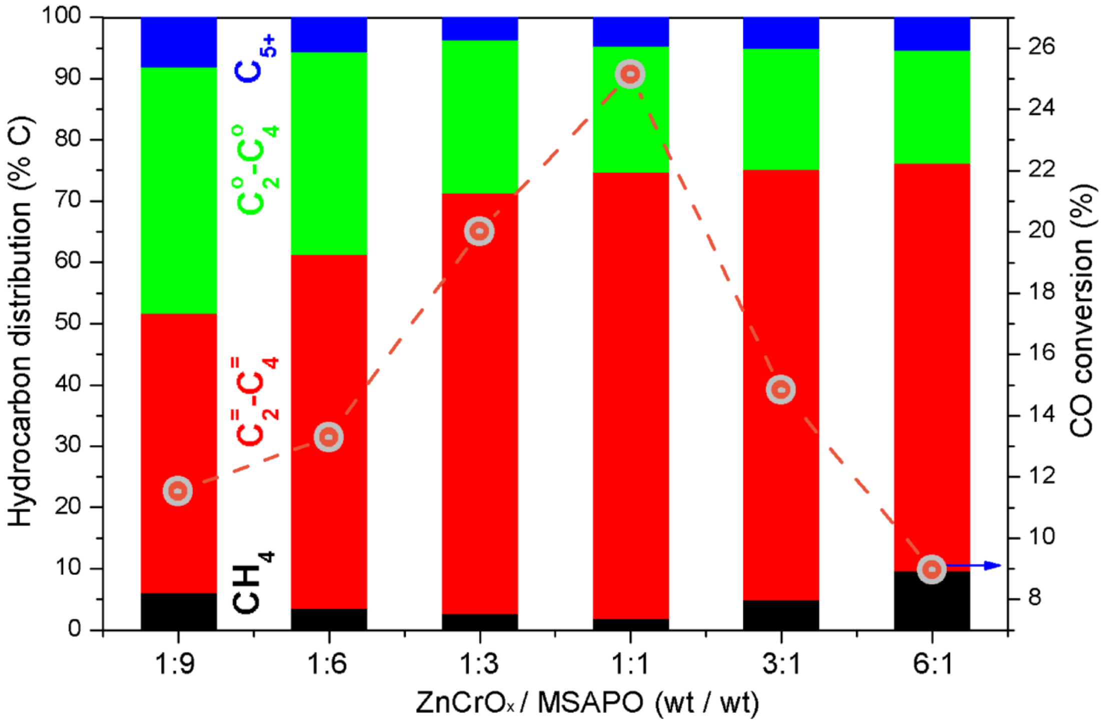

Fig. S4 Reaction results over composite catalysts as a function of the mass ratios of $ZnCrO_x$/MSAPO in the composite catalysts, which lend further support to the hypothesis that the composite catalyst is bifunctional and that the reaction involves intermediate transport in gas phase. Reaction conditions: 400 ℃, 2.5 MPa and GHSV 5143 ml/h g_{cat}.

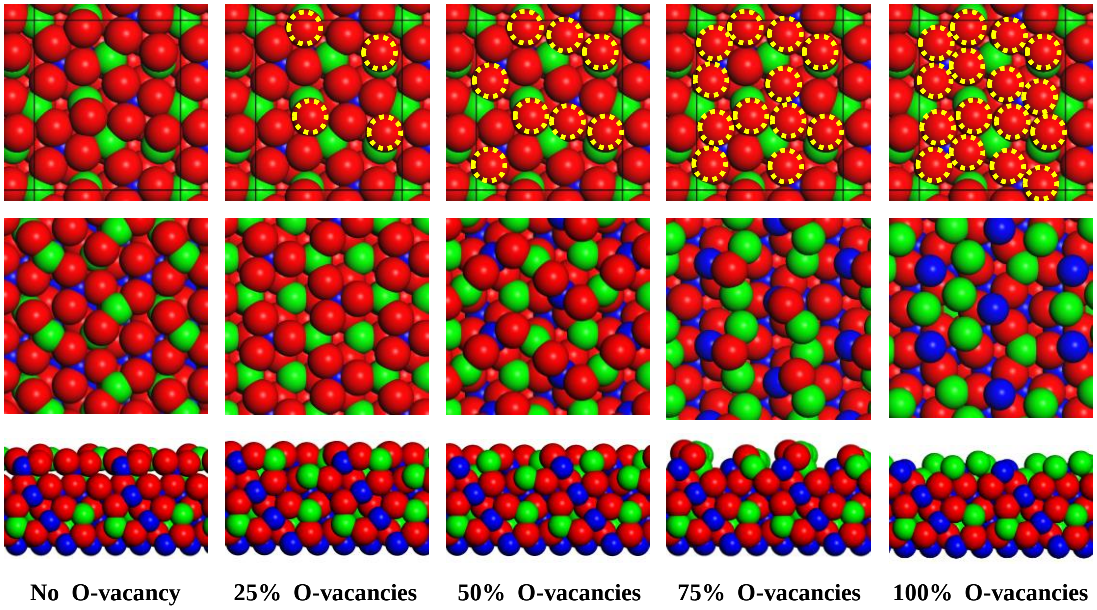

Fig. S5 Optimized models for spinel $ZnCr_2O_4$ containing different concentration of surface oxygen vacancies of 0%, 25%, 50%, 75% and 100% O at the first atomic layer, respectively. The top row shows the surface before optimization with the removed oxygen atoms marked in yellow dashed circles; the second row and the third row show the surface and side views of the optimized spinel $ZnCr_2O_4$ surface containing vacancies. Zn: green; Cr: blue; O: red.

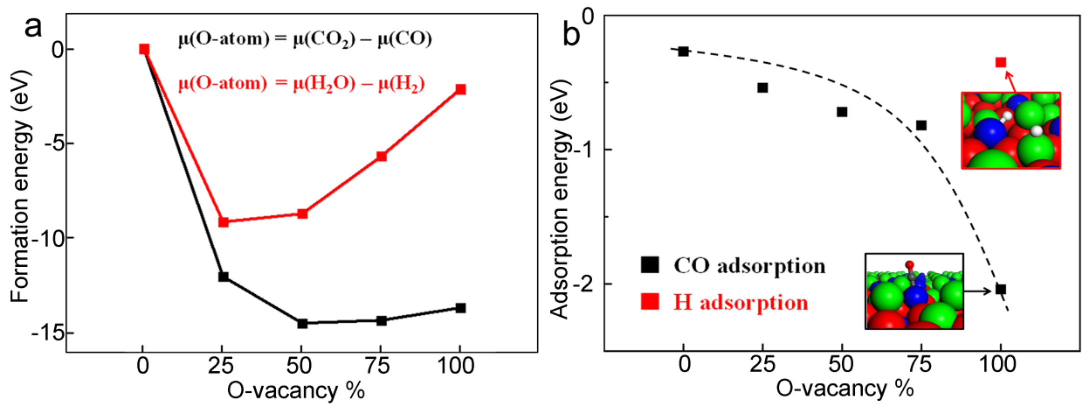

Fig. S6 (a) DFT calculations show that spinel $ZnCr_2O_4$ (111) surface is reducible and reduction by CO is much more energy favorable than by H_2, which leads to a partially reduced surface containing various concentrations of oxygen vacancies. The chemical potential of oxygen on the surface is assumed to be directly related with that of CO and CO_2 (black) for reduction by CO, and H_2O (red) for reduction by H_2, respectively. (b) Calculated adsorption energies for CO (black squares) and H atoms (red square) on the spinel $ZnCr_2O_4$ (111) surface with different concentration vacancies. The optimized CO and H_2 adsorption configurations on the surface with 100% O vacancies at the first atomic layer are shown in the insets. Zn: green; Cr: blue; O: red; H: white; C: grey.

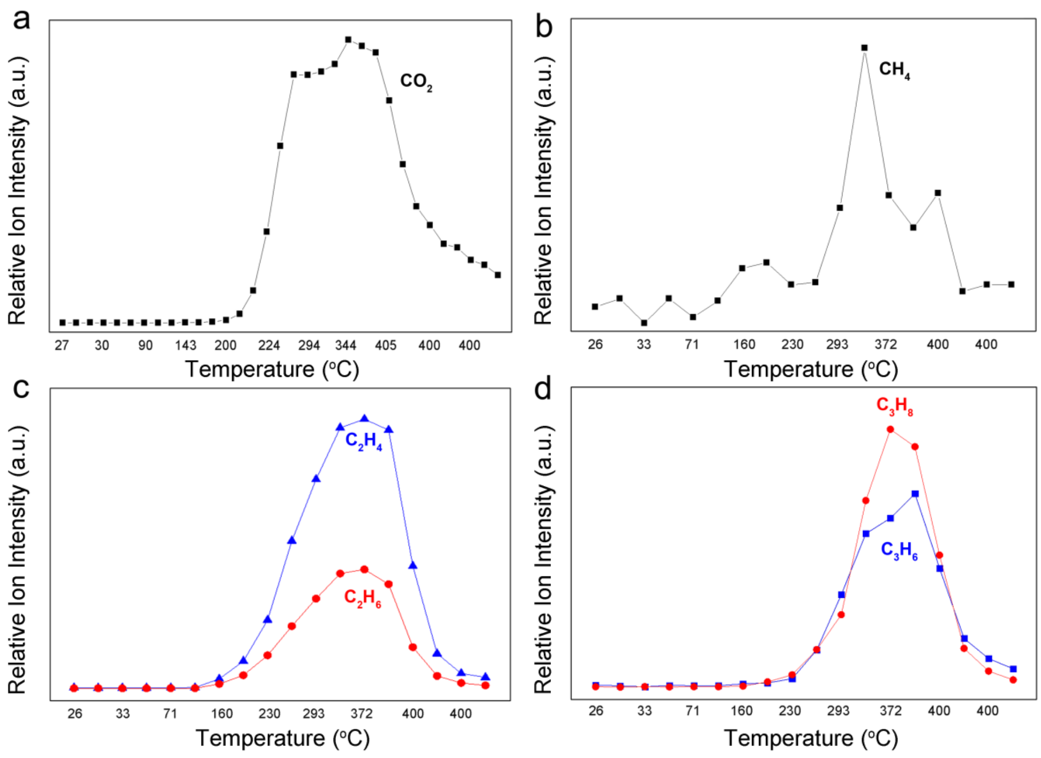

Fig. S7 (a) Temperature-programmed-redution of ZnCrO$_x$ by CO, with the effluent monitored by SVUV-PIMS. The results show that CO_2 was released with the temperature increasing. Extended exposure to CO at 400 ℃ leads to dissociation of CO forming CO_2 and surface *C species, evidenced by in situ NAP-XPS. The CO_2 yield declines with time due to increasing surface coverage of *C species. (b)–(d) Subsequent introduction of H_2 leading to formation of hydrocarbons due to hydrogenation of surface *C species. Hydrocarbon signals with *m/z* up to 126 are detected and those containing up to 3 carbon atoms are shown here as examples.

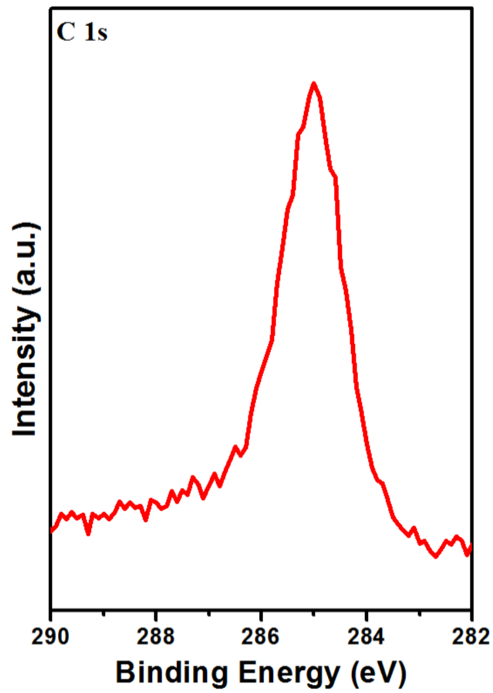

Fig. S8 In-situ NAP-XPS C1s spectrum of $ZnCrO_x$ recorded after evacuation in UHV at 300 ℃ following exposure to CO, indicating that this species is surface *C species from dissociated CO rather than surface adsorbed CO. The experiments were carried out at the Advanced Light Source, Berkeley.

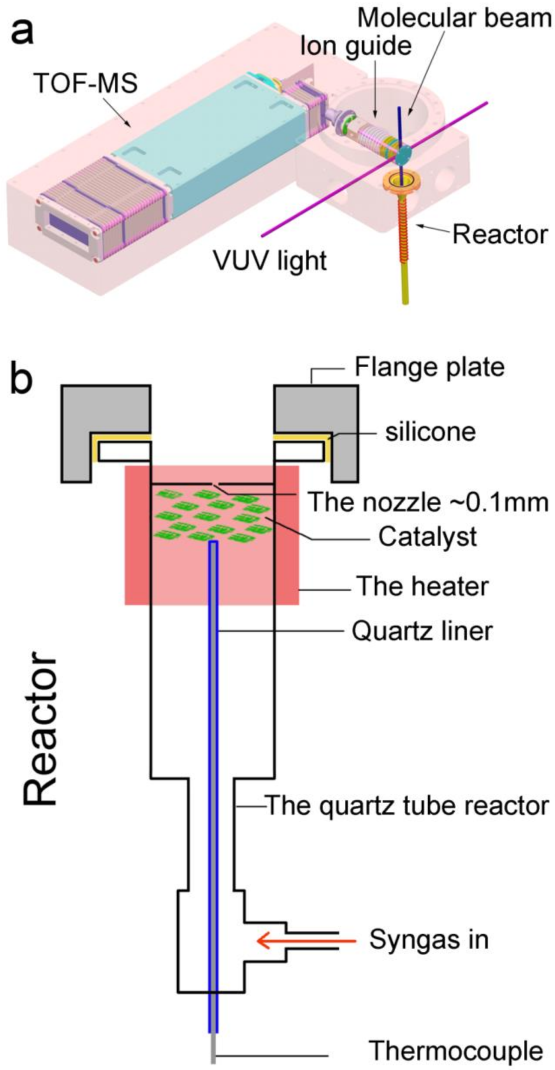

Fig. S9 Schemes for (a) the SVUV-PIMS system and it is connected with a catalytic reactor. (b) the catalytic reactor. The experiments were carried out at the combustion beamline of the National Synchrotron Radiation Laboratory at Hefei, China.

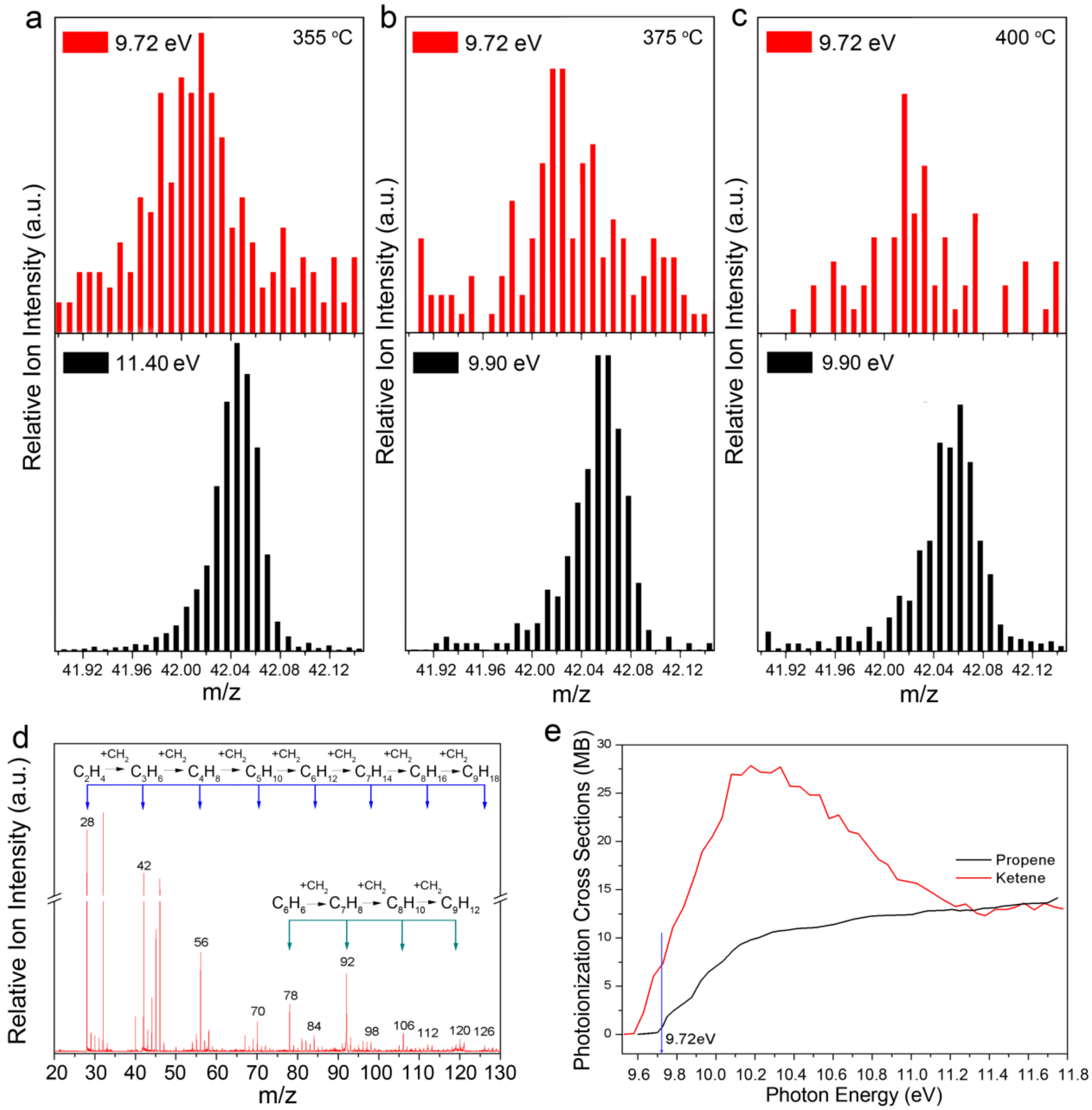

Fig. S10 In situ SVUV-PIMS study of syngas conversion over $ZnCrO_x$. (a) Signals of *m/z* = 42.01 (ketene) and *m/z* = 42.05 (propene), detected at 355 ℃ and different photon energies (hυ = 9.72 and 11.40 eV, respectively). (b) Signals of ketene and propene, detected at 375 ºC, and hυ = 9.72 and 9.90 eV, respectively. (c) Signals of ketene and propene detected at 400 ºC, and hυ = 9.72 and 9.90 eV, respectively. Although the intensity of ketene declines with the increasing temperature likely because of its higher reactivity, it is still discernible at 400 ℃. The *m/z* = 42.01 is unambiguously attributed to ketene CH_2CO considering the m/z ratio and the ionization energy. (d) A full spectrum collected at 355 ºC and hυ = 11.40 eV. Note that the intensity of ions is proportional to the concentration and the photoionization cross section. In addition to ketene and stable products of hydrocarbons, methanol (*m/z* = 32, ionization energy 10.84 eV) with a small amount of the dissociated product methoxyl group (*m/z* = 31) are also detected at this high hυ. However, over the composite catalyst the reaction does not likely go via methanol or methoxyl group because feeding methanol directly to the composite catalyst lead to dramatically dropping yield of $C_2^{=}-C_4^{=}$ down to 3% within 22 h in contrast to the stable performance in OX-ZEO as long as 650 h. MSAPO alone as a MTO catalyst also deactivates within 30 h even at a methanol partial pressure of 5–20 mbar. Furthermore, there is very little water produced in OX-ZEO process in contrast to MTO. (e) Absolute photoionization cross sections for propene (23) and ketene (24).

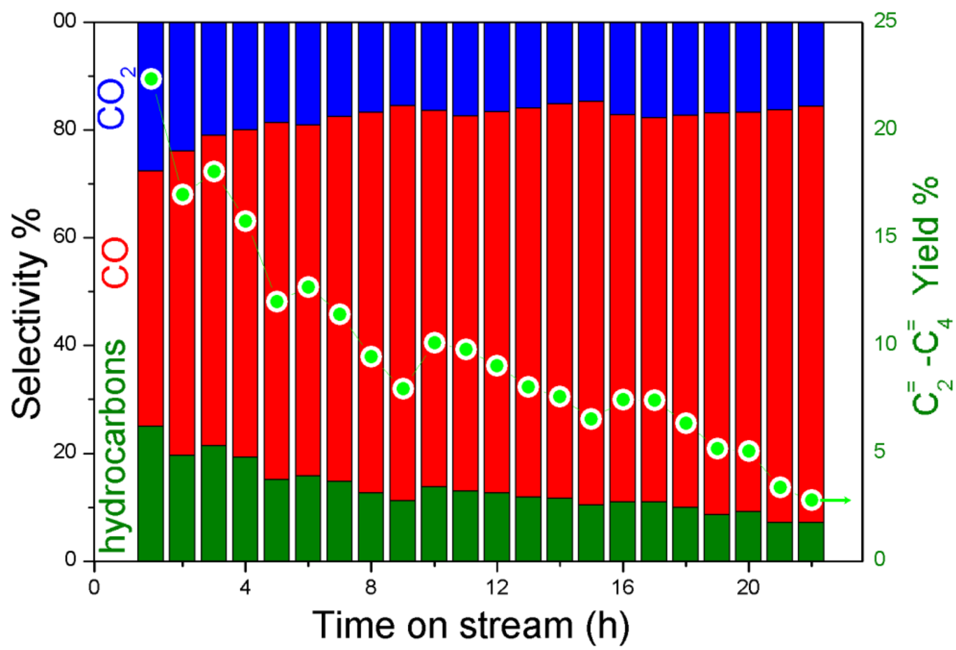

Fig. S11 Reaction results over the composite catalyst by feeding methanol (50 mbar in He) as the reactant at 400 ℃, WHSV of 0.5 h^{-1} and atmospheric pressure. The green filled circles represent the yield of $C_2^=$–$C_4^=$ and filled bars represent the selectivities of different products. A high selectivity of CO indicates that methanol mainly decomposes while the rest methanol goes through MTO reaction. However, the $C_2^=$–$C_4^=$ yield drops dramatically within 22 h and the selectivity to hydrocarbons also decreases quickly, indicating fast deactivation of the composite catalyst in MTO in contrast to the stable performance in syngas conversion in a 650 h test.

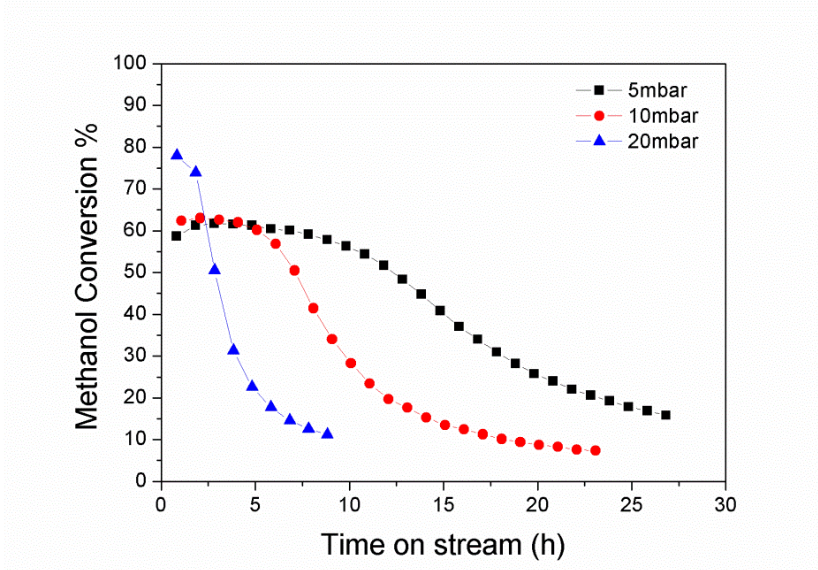

Fig. S12 Reaction results over the MSAPO catalyst by feeding methanol as the reactant at 400 ℃ at varying methanol partial pressures, 5, 10 and 20 mbar using He as the carrier gas. The amount of methanol feeding was kept at a value to achieve the same olefin yield as in OX-ZEO process assuming 100% conversion of methanol. The results show quick deactivation of the catalyst around 20 h at 10 mbar and within less than 30 h at an even lower partial pressure of 5 mbar converting 0.18 mol C/g_{cat} before deactivation. This is in contrast to the stable performance as long as 650 h in our OX-ZEO process and 9.6 mol C/g_{cat} was converted within this period of time. Furthermore, methanol conversion far lower than 100% indicates that not all olefins are produced from methanol. These results suggest that methanol is not the dominating reaction pathway. There should exist another reaction pathway.

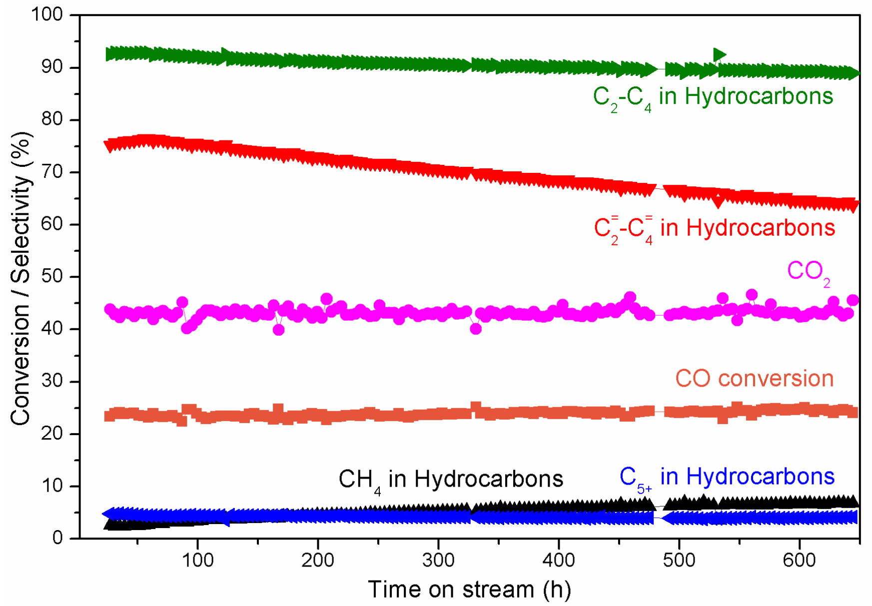

Fig. S13 A stability test for 650 h under condition of H_2/CO = 2.5, 400 ℃ and 5143 ml/h g_{cat}. This catalyst has an oxide/zeolite mass ratio of 1/3.

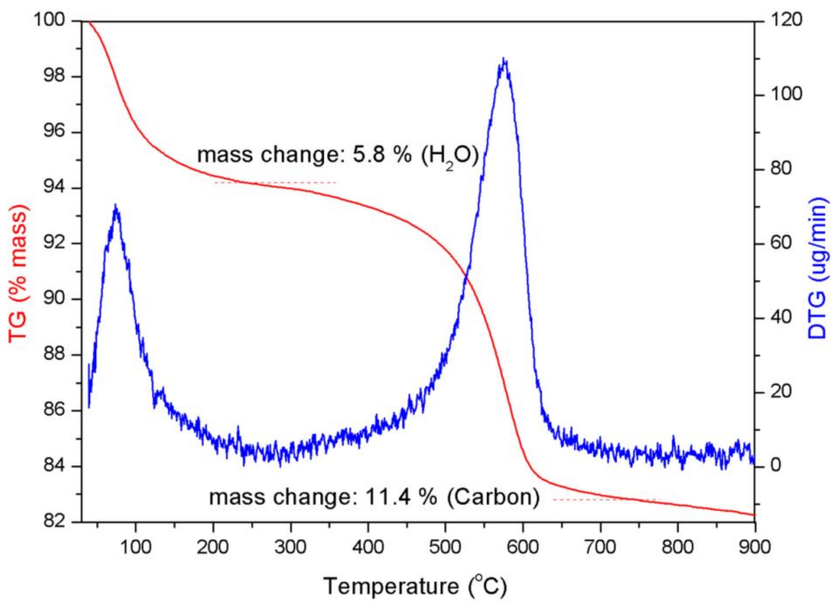

Fig. S14 TG & DTG analysis of the composite catalysts after reaction tests of 650 h in Fig. S13.

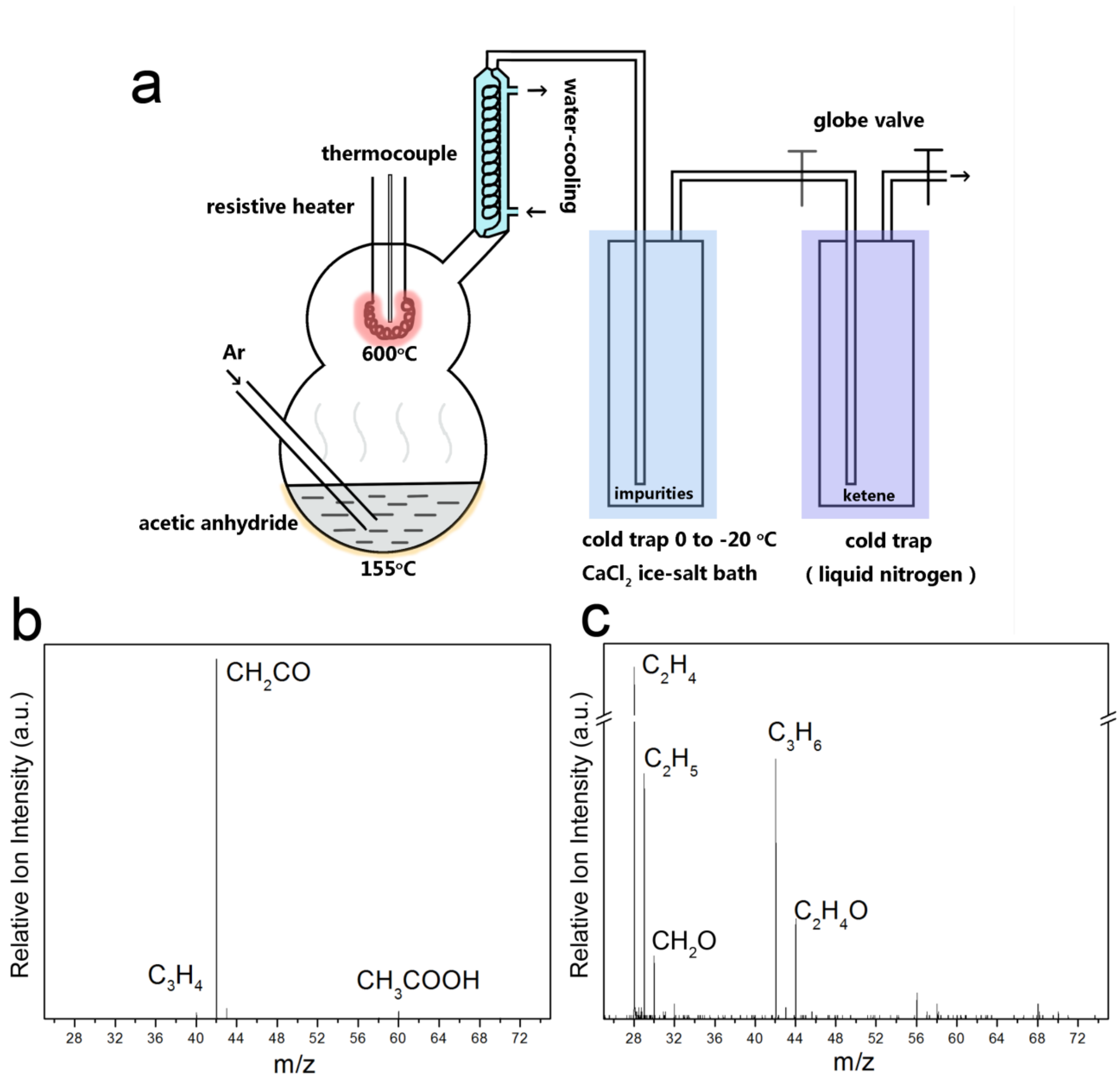

Fig. S15 (a) Experimental setup for synthesis of ketene from acetic anhydride via pyrolysis following the reported procedure (24, 28); (b) Blank experiment with ketene fed to the empty reactor in the absence of the catalyst. The strong peak of *m/z* = 42.01 corresponds to ketene while the other smaller peaks are attributed to impurities in ketene. (c) Results with ketene fed to the MSAPO catalyst. Reaction temperature was 400 ℃ with the effluents monitored by the online SVUV-PIMS and the spectra were recorded at 10.70 eV.

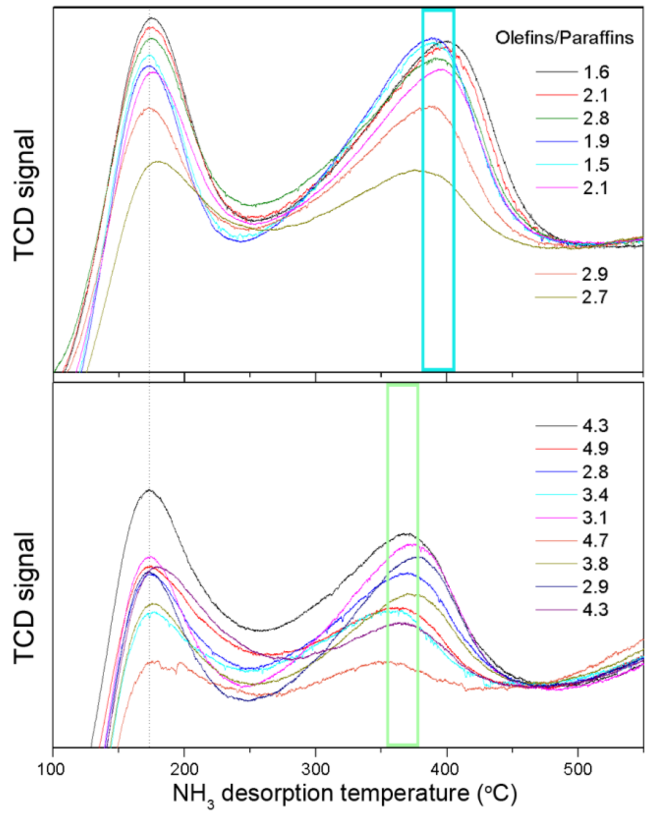

Fig. S16 NH_3-TPD profiles of a series of MSAPO samples, which lead to products with varying selectivity ratios of olefins to paraffins ($C_2^{=}$−$C_4^{=}$)/(C_2^{o}−C_4^{o}). The selectivity ratios of olefins/paraffins are listed in the figure, demonstrating importance of medium strength acidity on the olefin selectivity. Reaction conditions: 400 ℃, 2.5 MPa and GHSV 5143 ml/h g_{cat}.

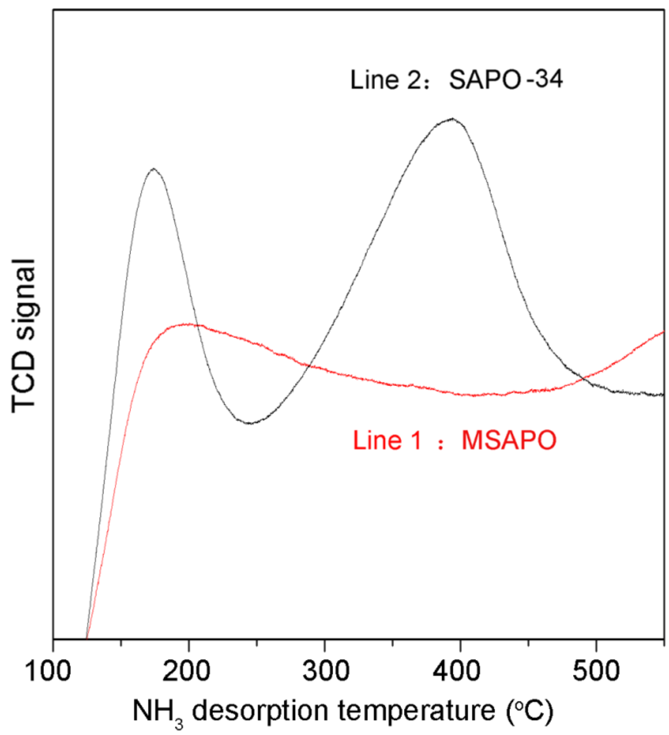

Fig. S17 NH_3-TPD profiles for a MSAPO (Line 1) and a commercially available microporous SAPO-34 (Line 2, purchased from Nankai University Catalyst Company, China). Line 1 represents a MSAPO with almost no medium strength acidic sites and the composite catalyst containing this MSAPO gives a similar product distribution as the $ZnCrO_x$ only catalyst. Line 2 represents SAPO-34 with a peak maximum of medium acidity sites at ~394 ℃, which gives a C_2^o–C_4^o paraffin selectivity over 50% and $C_2^=$–$C_4^=$ selectivity of 43% (Table S3). The results indicate that both the hierarchical pore structure and the medium strength acidity play important roles in determining the selectivity of olefins. By contrast, composite catalysts containing metal oxides (such as Cr_2O_3-ZnO, CuO-ZnO) and a variety of conventional zeolites (ZSM-5, Y, β) yielded mainly dimethyl ether (10), liquefied petroleum gas (C_3^o–C_4^o) (11-13) or gasoline (14, 15).

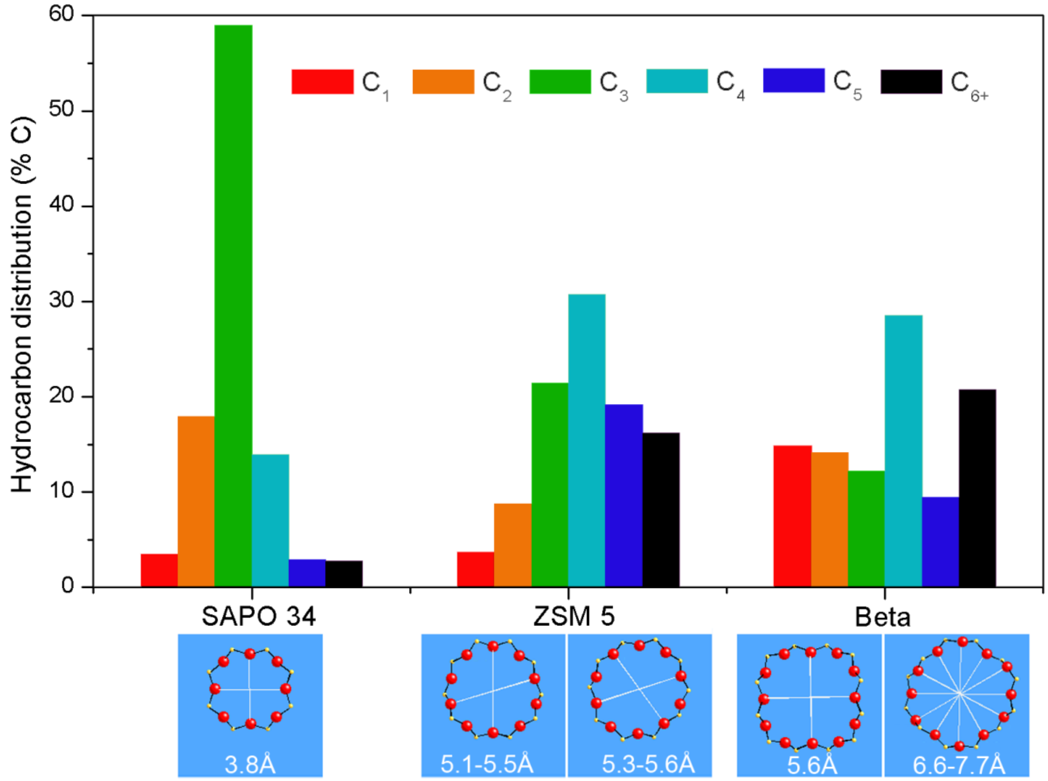

Fig. S18 Hydrocarbon distribution over composite catalysts containing different zeolites including the commercially available SAPO-34, ZSM-5 and Beta zeolites. Reaction conditions: 400 ℃, 2.5 MPa and GHSV 5143 ml/h g_{cat}.

Table S1. Surface areas and pores of MSAPO and $ZnCrO_x$ obtained from N_2 adsorption-desorption at 77 K.

Sample	BET specific surface area	Micro-area	Meso-area	Micropore volume	Pore volume	Average mesopore diameter
	m^2/g	m^2/g	m^2/g	cc/g	cc/g	nm
$ZnCrO_x$	98	0	98	0	0.28	~12
MSAPO	365	151	214	0.07	0.35	~7.6

Table S2. Detailed reaction results in Fig. 2A.

Space Velocity	**CO Conversion %**	**CO_2 Selectivity %**	**Methanol Selectivity %**	**Hydrocarbon distribution, %**			
				CH_4	$C_2^=–C_4^=$	$C_2^o–C_4^o$	C_{5+}
			Mode 1				
1285	7	34	1	53	19	19	9
3857	5	25	2	47	26	18	9
7714	4	29	2	48	30	13	9
			Mode 2				
1285	9	37	a	26	23	46	5
3857	6	35	a	20	40	35	5
7714	4	48	a	16	54	26	4
			Mode 3				
1285	25	41	a	8	45	42	5
3857	15	41	a	8	57	31	4
7714	10	42	a	7	65	24	4
			Mode 4				
1285	35	40	a	3	69	20	8
3857	23	41	a	3	76	15	6
7714	17	41	a	2	80	14	4

a: the selectivity to methanol was below 0.5%C and was not reported in the list.

Table S3. Detailed reaction results of fig. S17.

	CO conversion %	**CO_2 Selectivity %**	**Methanol Selectivity %**	**Hydrocarbons distribution %**			
				CH_4	$C_2^{=}$–$C_4^{=}$	C_2^{o}–C_4^{o}	C_{5+}
Line 1 MSAPO	10	32	1	34	37	18	11
Line 2 SAPO-34	32	39	a	4	43	50	3

a: the selectivity to methanol was below 0.5%C and was not reported in the list.

Table S4. The equilibrium concentration of methanol and ketene from thermodynamic calculations using HSC-9.0 software under 400 ℃, 25 bar, with 1.5 kmol H_2 and 1 kmol CO as the starting compounds.

Components of the effluents	Methanol	ketene
Case 1: CO, H_2, CH_3OH (methanol)	1137 ppm	—
Case 2: CO, H_2, CH_2CO (ketene), CO_2	—	8740 ppm
Case 3: CO, H_2, CH_3OH, CH_2CO, CO_2	1089 ppm	8727 ppm
Case 4: CO, H_2, CH_3OH, CH_2CO, H_2O	1116 ppm	3267 ppm
Case 5: CO, H_2, CH_3OH, CH_2CO, H_2O, CO_2	1085 ppm	9270 ppm

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